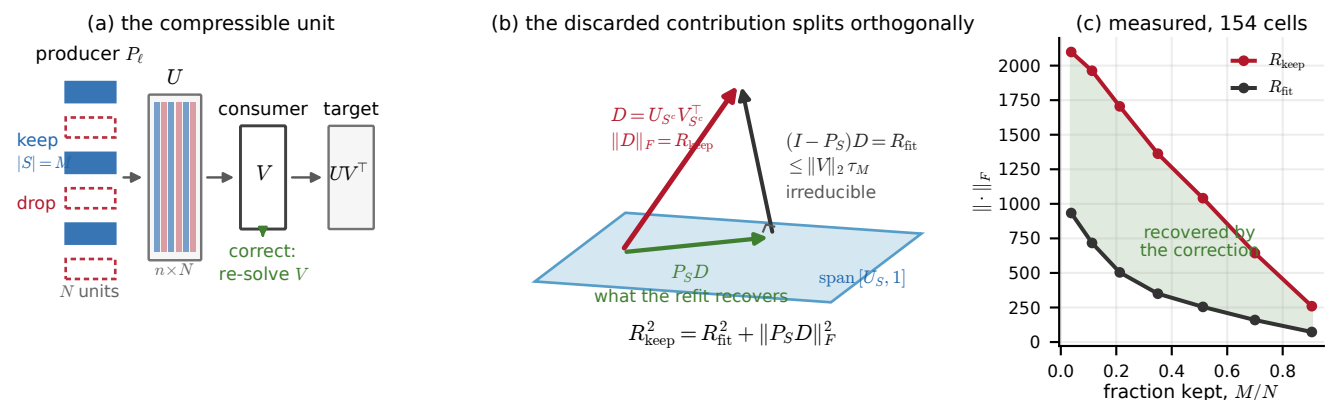
(a) the compressible unit
producer P_ℓ
U
consumer
target
keep
$|S| = M$
drop
V
$UV^\top$
correct:
re-solve V
$n \times N$
N units
(b) the discarded contribution splits orthogonally
$D = U_{S^c} V_{S^c}^\top$
$\|D\|_F = R_{\text{keep}}$
$(I - P_S)D = R_{\text{fit}}$
$\leq \|V\|_2 \tau_M$
irreducible
$P_S D$
what the refit recovers
span $[U_S, 1]$
$R_{\text{keep}}^2 = R_{\text{fit}}^2 + \|P_S D\|_F^2$
(c) measured, 154 cells
R_{keep}
R_{fit}
recovered by
the correction
$\|\cdot\|_F$
2000
1750
1500
1250
1000
750
500
250
0
0.0
0.2
0.4
0.6
0.8
fraction kept, M/N